

March 15, 2025

AI Action Plan
Attn: Faisal D'Souza, NCO
2415 Eisenhower Avenue
Alexandria, VA 22314, USA

Reference: Response to the White House Office of Science and Technology Policy RFI on
Sustaining and Enhancing America's AI Leadership

Dear Mr. D'Souza:

The race to dominate artificial intelligence (AI) represents the defining technological challenge of our time. AI's rapid advancements have catalyzed scientific progress, reshaped industries, and reinforced its national security significance. Competing nations are making strategic investments in sovereign AI capabilities, narrowing key gaps with the United States. To maintain global leadership, the U.S. Government must make targeted, bold investments to sustain and advance our AI leadership.

As the U.S. private sector spearheads cutting-edge AI innovations, the federal government must play a pivotal role in steering these advancements to enhance national security, accelerate scientific progress, and transform the work of federal agencies. Leveraging existing laboratories and facilities, a world-class scientific workforce, and the nation's most advanced computing resources can help ensure the United States remains the global superpower in AI-driven science, manufacturing, energy, national security, and economic competitiveness. This approach maximizes efficiency, safeguards taxpayer dollars, and accelerates AI innovation.

The U.S. Government has a strong track record of mission-driven collaborations with industry, ensuring the rapid translation of advanced computing research, development, and deployment into real-world applications. A model example is the Exascale Computing Initiative (ECI), which led to the successful co-design, co-development, and co-funding of supercomputing and AI technologies. This initiative resulted in AI-enabled supercomputers that are 1,000 times more powerful and 100 times more energy-efficient than the prior generation, while contributing to critical semiconductor advancements that power today's leading AI models. As a long-standing technology partner to the U.S. Government, HPE has played an integral role in supporting these mission-critical computing initiatives. By providing AI-driven infrastructure that advances national security, scientific discovery, and cutting-edge research, HPE continues to serve as a trusted strategic partner for government-funded AI programs. Ensuring continued federal investment in AI-driven computing will be essential to reinforcing U.S. technological dominance and accelerating AI innovation.

The ECI demonstrates the power of strategic federal investment that leverages public and private partnerships. As a result of this initiative, the U.S. now owns the #1, #2, and #3 most powerful supercomputers in the world; systems supporting national security and cutting-edge

science. Versions of these supercomputers are now supporting mission-critical applications across the Department of Defense, Intelligence Community, Department of Energy, and other agencies, including at NOAA for severe weather forecasting. Expanding this model of success to a nation-scale AI initiative can accelerate progress and reinforce U.S. technological dominance.

In determining and prioritizing areas of AI research, development, and demonstration requiring federal leadership and investment to maintain U.S. technological dominance, we recommend focusing on the following key areas:

- **Data:** The U.S. Government is the world's leading generator of both classified and unclassified scientific data. Harnessing this vast repository for AI applications can drive breakthroughs in healthcare, energy science, manufacturing, and national security, and beyond.
- **Computing:** For decades, the U.S. Government has built and operated the world's most powerful and energy-efficient supercomputers, providing strategic assets for scientific discovery and national defense across the nation. These high-performance computing resources and supporting ecosystems can provide the foundation upon which safe and trustworthy AI capabilities can be built and expanded.
- **Government Readiness:** The U.S. national laboratories employ tens of thousands of experts across physics, chemistry, biology, materials science, and computer science, including nearly one-third of all Ph.D.-level scientists and engineers in the country. This unparalleled talent pool is vital for advancements in AI and for the future of science and engineering.

To sustain and enhance America's AI leadership, the federal government must take decisive action in guiding AI development through strategic investments in data, computing, workforce, and public-private collaboration. By building upon our existing strengths, we can secure the future of AI leadership and ensure continued scientific and economic prosperity for the nation.

Thank you for the opportunity to provide input. I welcome further discussion on these recommendations.

Please contact me, Mark Guiton, with any clarification questions at [REDACTED] or [REDACTED] (cell).

Sincerely,

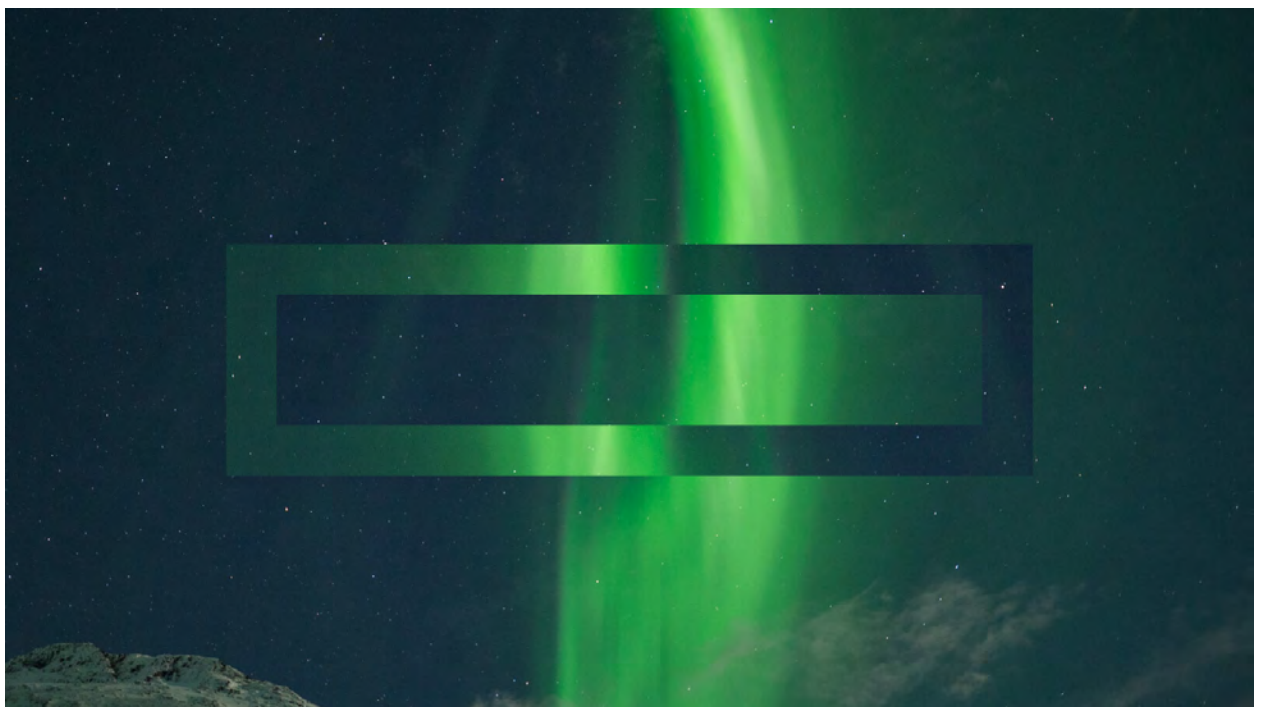
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The White House

White House AI Action Plan

Response to Request for Information on the Development of an
Artificial Intelligence (AI) Action Plan



Technical Volume

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1. Introduction

The rapid advancement of AI is transforming numerous sectors, fueling innovation, reshaping high-performance computing (HPC), and driving IT modernization. To sustain and strengthen American AI leadership while fostering continued innovation, the United States must adopt a comprehensive strategy.

Investing in a large-scale system infrastructure for AI, including networking and storage, is crucial to harnessing emerging AI technologies. This includes addressing the challenges of managing large data volumes while ensuring data security and collaborative practices. Adopting industry standards and avoiding vendor lock-ins will foster competitiveness and innovation.

Data centers are the backbone of AI innovation, providing the necessary infrastructure for computational power and secure data processing. Strategic data center locations, with availability of multiple energy sources, are crucial for national security and data sovereignty. Because the increasing adoption of AI drives up energy demands, investing in energy-efficient technologies such as liquid cooling and dynamic power management solutions is essential for managing energy consumption and optimizing performance at a given energy level.

AI model development faces challenges related to computational costs and data quality. Investing in efficient algorithms and fostering public/private partnerships enhances collaboration and innovation. Promoting open-source frameworks and clear guidelines for data usage can amplify the community contributions and open the door for new AI techniques and applications. As part of that effort, government agencies must modernize their IT infrastructure to be ready for AI. Automating IT operations, upgrading servers, networking and storage, and developing an AI-ready workforce are key steps. Transparency and robustness in AI systems are critical and investing in explainability methods will expand the applications domains of AI to mission critical areas. Cybersecurity also requires a multi-faceted approach to secure AI systems throughout their lifecycle.

In conclusion, these actions will enable the United States to sustain and enhance its AI leadership, promote U.S.-driven AI standards, and ensure that innovation thrives while maintaining national security and economic competitiveness.

HPE's response to the White House Office of Science and Technology Policy RFI on Sustaining and Enhancing America's AI Leadership considers statements made in the following executive actions:

- <https://trumpwhitehouse.archives.gov/presidential-actions/executive-order-maintaining-american-leadership-artificial-intelligence/>
- <https://trumpwhitehouse.archives.gov/wp-content/uploads/2019/11/National-Strategic-Computing-Initiative-Update-2019.pdf>
- <https://trumpwhitehouse.archives.gov/wp-content/uploads/2020/02/American-AI-Initiative-One-Year-Annual-Report.pdf>
- <https://www.nitrd.gov/pubs/National-AI-RD-Strategy-2019.pdf>

2. HPE's Policy Recommendations

HPE demonstrates strong technical leadership in AI infrastructure, HPC, energy-efficient AI systems, and federated AI model training. HPE provides input on the highest priority policy actions that should be in the AI Action Plan.

HPE is focused on being a strategic AI policy leader on governance, workforce, cybersecurity, and regulatory policies. HPE is capable and eager to continue working closely with the U.S. Government to secure federal partnerships, influence AI policy direction, and lead and collaborate on government AI projects.

2.1 Computing

The pervasiveness of AI has been the largest disruption to computing in the last five years. AI technology has driven the direction of hardware and software technologies and transformed the way scientists perform their work. Engineers developing and using AI models will continue to need high-performance supercomputers to solve some of the world's most challenging problems. By taking a leadership role in investing and building at-scale AI system infrastructure, including networking and storage, the U.S. can harness hardware and software capabilities and technologies being developed across the world.

Challenges: The next phase of AI development will have to focus on providing a robust, scalable, reliable, and secure infrastructure to build massive AI supercomputers and run complex and diverse AI workflows. The challenges in this space are numerous: managing large data amounts with guaranteed security, building efficient and open-standards networking, enabling scalable power-management solutions, and providing portable software stacks.

There have been several specialized and purpose-built compute silicon offerings from AI startups that have shown significant speedups on a handful of applications in testbed environments. Currently, these accelerators tend to be deployed independently, and workflows and surrogate models are challenged to seamlessly leverage the capabilities of multiple accelerator types. Investment in holistic system architecture and making system-level heterogeneity work well from a hardware and software perspective is critical to making effective use of such specialized accelerators.

Opportunities: The massive needs of AI at every scale (for both training and inference) require building compute infrastructure that scales on a first-of-its-kind with Ethernet-compatible network fabric integrated with pervasive security. A robust standards-driven network interconnect infrastructure is the foundation to support performance and productivity improvements – enabling new modes of scientific discovery from edge to extreme scale systems. This vision requires investments that are aligned with cutting-edge technological advancements and supporting new industry-wide initiatives, such as the Ultra Ethernet Consortium (UEC) or Ultra Accelerator Link (UAL).

A robust supply chain is essential for the needs of AI hardware and infrastructure. The U.S. can provide significant leadership in this space by supporting domestic industries and companies that can help ensure a healthy supply chain.

High-performance storage and data management are critical to supporting large language models. AI for commercial purposes (for example, manufacturing and science) depends on a community that can maintain the privacy and security of mission-critical data sets but shares knowledge and learnings on new hardware, algorithms, standardized tools, and evaluation standards. Collaborating, leading, and investing in such efforts are vital to the U.S. maintaining leadership and influence in AI across the world.

Recommended actions include:

- Increase investment in AI chip manufacturing and domestic semiconductor leadership.
- Increasing U.S. Government investment in system, storage, and interconnect network infrastructure to support heterogeneity in hardware and software technologies.
- Adopting industry standards and initiatives to avoid vendor lock-ins and enable competitiveness across the ecosystem.
- Investing today to better prepare for the shift toward AI-driven science that is expected in the future by supporting consortiums like the UEC, UAL, and Trillion Parameter Consortium (TPC).

2.2 Data Centers

Data centers are the backbone of AI innovation and play a pivotal role in ensuring the United States maintains technological superiority in the global AI race. As AI models become increasingly complex, requiring vast computational power and secure, high-speed data processing, the development of scalable and resilient data center infrastructure is critical. These facilities not only house the hardware necessary for AI supercomputing but also enable seamless integration of specialized accelerators, such as GPUs and emerging AI-specific chips, that are essential for optimizing machine learning workflows. The location of data centers is also critical for national security and data sovereignty. AI operations rely on sensitive data, making the choice between co-location, on-premise, or cloud-based infrastructure strategic. With China rapidly advancing its AI capabilities, securing a robust, efficient, and secure AI infrastructure is a matter of national security and economic competitiveness.

Recommended actions include:

- Leading open standards in networking, scalable power management, and secure data access can cement dominance in AI and shape the future of global technological leadership.
- Investing in new and existing AI-ready data centers, such as the Department of Energy (DOE) facilities, will ensure that the U.S. can support next-generation advancements in quantum computing, autonomous systems, and high-performance computing.

2.3 Energy Consumption and Efficiency

AI has seen a jump in usage across many different sectors from the service industry to advancing manufacturing and research. This increased usage across multiple industries is driving power demands across the whole data center industry. The energy consumption and, therefore, costs associated with operating data centers, are predicted to double or even triple in the next four years mainly due to the use of AI.

Challenges: The unprecedented energy consumption increase driven by adoption of AI has the potential to drive energy costs up and to put pressure on the country's energy capacity and availability. Chip and system power density increases create challenges in removing the waste heat. Being able to operate AI systems economically and stand-up additional power capacity quickly without interruptions will be key competitive factors in the race to AI leadership.

Opportunities: Tighter integration of data centers with energy supply grids will improve reliability and accelerate the expansion of AI capacity while decreasing costs. Carefully managing system-level power consumption will enable data centers to recapture stranded power capacity and enable the U.S. to generate more AI capacity while limiting additional energy supply infrastructure costs. Additionally, investments in power and energy management software can help to optimize performance, reduce operating costs, and increase the overall return on investment. Investing in high-performance and energy efficient cooling technologies from the chip, system, and data center will be necessary to maintain a leadership position in AI. Liquid cooling has been shown to be more efficient than air cooling, enabling higher compute density and leading to reduced overall costs.

Recommended actions include:

- Encouraging adoption and innovation of liquid cooling technologies at all levels (AI devices, systems, and data centers).
- Supporting innovations that decrease investment costs, reducing permitting and deployment time, and maximizing AI capabilities.
- Investing in dynamic system power and energy management software.
- Investing in digital twin technology to optimize the design and operation of multiple sub-systems throughout the data center, maximizing the overall return on investment by balancing performance, capital expenses, and operating costs.
- Supporting tighter integration of data centers with energy supply grids, microgrids, and behind-the-meter power generation to accelerate AI capacity and capability expansion.

Pursuing these recommended actions will enable the U.S. to develop more AI capabilities at a similar investment level than our competitors, faster, and leading to the highest AI capabilities per investment.

2.4 Data

Challenges and opportunities exist in the gathering, validation, security, and use of data in the areas of model and open-source development.

2.4.1 Model Development

AI model development has advanced rapidly, driven by innovations in deep learning, reinforcement learning, reasoning algorithms, and transfer learning. Current state-of-the-art foundation models have shown remarkable capabilities in natural language understanding and generation, and reasoning. However, these advancements come with significant computational and energy costs. Data repositories play an essential role in AI model development by providing the vast amounts of data necessary to train, test, and validate AI models. These repositories, when managed and curated effectively, ensure that high-quality, diverse, and relevant datasets are available for AI research and development. This accessibility accelerates innovation, fosters collaboration, and enhances the reproducibility of scientific research.

Challenges include the need for vast amounts of labeled data, high computational costs, and the difficulty of ensuring model robustness and fairness. Explainability remains a significant hurdle, as complex models often operate as "black boxes," as well as ensuring the quality and consistency of the data. Different data sources may have varying formats, standards, and levels of detail, which can complicate their integration. Additionally, maintaining data privacy and security is crucial, especially for sensitive data.

Opportunities include investing in more efficient algorithms, heterogeneous hardware acceleration, and resource-constrained approaches to reduce data dependency and computational costs.

Emphasizing trustworthy AI principles prioritizes robustness and transparency. Public data repositories, such as those managed by the Department of Energy and other federal agencies, are crucial for supporting AI research. These agencies, such as DOE's Office of Science, control valuable data that is not available elsewhere. Tokenizing those valuable data sets to generate AI-ready formats fosters innovation across multiple domains and facilitates the creation of AI foundation models for science. These repositories can store extensive datasets from a variety of scientific domains, making them accessible to researchers and developers. For example, the National Oceanic and Atmospheric Administration (NOAA) provides valuable long-term weather data that is indispensable for developing AI models in environmental science.

Recommended actions include:

- Encouraging adoption of voluntary risk-based approaches for risk assessment.
- Fostering public/private partnerships for research and investment. There is a significant opportunity to enhance collaboration between public and private sectors, as well as among different scientific disciplines. By implementing standardized data governance practices and leveraging technologies such as AI for data curation and management, or generation of synthetic data, the initiative can improve the usability and reliability of public data repositories.
- Adopting flexible regulations that do not stifle innovation and support the development of scalable and robust AI systems. This may include standardizing AI model benchmarks and quality criteria for consistent evaluation.

- Launching an AI Data Initiative is necessary to streamline and enhance the availability and accessibility of these datasets. Such an initiative would focus on creating a federated data infrastructure that ensures data quality, security, and ease of access. This infrastructure would facilitate the integration of datasets from various sources, making it easier for researchers to find and utilize the data they need.
- Creating a Federated Data Infrastructure that integrates datasets from multiple sources, ensuring they are easily accessible and usable for AI research. This infrastructure should support standardized metadata and interoperability across platforms.
- Supporting Data-Centric AI Research by shifting the focus from model-centric to data-centric AI research. This approach emphasizes the importance of high-quality data in developing robust and reliable AI models.
- Promoting Open Standards and Interoperability by advocating for the adoption of open standards and APIs to facilitate seamless data integration and sharing across different platforms and systems. This will enhance collaboration and innovation in AI research.

2.4.2 Explainability and Assurance of AI Model Outputs

Uses of AI in mission-critical applications highlight the critical importance of explainability and assurance in AI model outputs, and the need for transparency and robustness in AI systems. As AI technologies permeate various sectors, they necessitate the development of explainable and robust models, which involve creating AI Principles tailored to every specific mission. These principles guide the operationalization of commitments and specifications, ensuring that AI technologies are utilized correctly across products, processes, and partnerships.

For discriminative AI (e.g., computer vision), explainability techniques (such as LIME and SHAP) and various visualization tools are being developed to interpret complex AI models. These tools are crucial in domains where transparency and accountability are paramount, such as healthcare, manufacturing, defense, and finance.

For generative AI, evaluations have grown in importance and encourage transparency in large language models (LLMs). By establishing rigorous standards and benchmarks, these evaluations assess the robustness, fairness, and explainability of LLMs, thereby fostering trust and accountability in their outputs.

Challenges: Balancing model complexity and explainability is difficult. Ensuring that explanations are understandable and actionable for non-experts remains a challenge.

Opportunities: Investing in research to improve explainability methods and integrating them into AI development processes. This can enhance trust and adoption of AI systems in critical applications. The NIST AI Safety Institute plays a vital role in fostering the adoption of AI auditing practices to identify and mitigate potential model vulnerabilities. Aligning with NIST's initiatives, particularly in "red teaming" technologies for specific science models, will help verify model correctness and ensure safety.

Recommended actions include:

- Investing in technical solutions for explainability and red teaming of AI models, accelerating the work of NIST's AI Safety Institute.
- Using techniques such as reinforcement learning to develop models that can provide intuitive explanations.
- Fostering partnerships between academia, industry, and government to advance explainability research.

2.4.3 Open-Source Development

Open-source AI projects, such as TensorFlow, PyTorch, and Hugging Face's communities, have democratized access to AI tools and resources. These platforms enable a wide range of users to develop and deploy AI models in a portable way. Leveraging the rapidly growing AI offerings in the cloud is important, but equally important is ensuring that users maintain the freedom to move their developments across different systems. This requires establishing programming practices based on open standards that are independent of specific proprietary accelerators (only available behind a cloud service API).

In addition to open-source codes, open models (open-source and open-parameters) foster AI community collaboration, transparency, and scientific advancements. Opening foundation models for science can democratize access to cutting-edge technologies, foster collaboration, and accelerate scientific discovery. Initiatives like the Trillion Parameter Consortium (TPC) provide a fertile ground for research teams around the world to collaborate on important AI research challenges for science and create a focal point to compete with commercial AI that drives much larger business interests. When models are openly available, scientists can build upon each other's work, leading to rapid advancements and breakthroughs.

Challenges: Ensuring the security and quality of open-source AI models and maintaining community contributions and addressing intellectual property issues can also be challenging.

Opportunities: Enhancing collaboration through open-source platforms can accelerate innovation and adoption of AI technologies. Leveraging community-driven development can lead to more robust and diverse AI solutions. Open-source initiatives in fostering innovation and collaboration within the AI community are very important. The development and support of a broad-based open-source community dedicated to creating an innovation ecosystem around AI helps to seamlessly integrate edge-to-cloud-to-supercomputing AI. The open-source approach is crucial for enabling continuous integration, iterative development, and collaborative advancements in AI technologies. Providing open-source frameworks while recognizing standards bridge gaps in industry practices and ensuring that AI innovations are accessible and beneficial to a wider range of stakeholders is important. Moreover, public-private partnerships are key in promoting open-source software and preventing vendor lock-in. The establishment of an open infrastructure environment would support both AI hardware and software innovations from the broader community. By maintaining openness in data protocols, programming paradigms, and software libraries, the AI ecosystem can thrive through collective contributions and shared best practices. This approach not only enhances transparency and

interoperability but also democratizes access to AI technologies, allowing a broader base of U.S. researchers and developers to contribute and benefit from advancements in AI.

Recommended actions include:

- Promoting open-source frameworks and standards.
- Establishing clear guidelines for data usage and model development.

2.5 Government AI Readiness

Several challenges and opportunities exist for the U.S. Government spanning the areas of AI applications and their use, explainability and assurance of AI model outputs, cybersecurity, data privacy and security, national security and defense, research and development, innovation and competition, procurement, and international collaboration. HPE's assessment and recommendations leverage HPE's past and current leadership in AI infrastructure and national security applications.

2.5.1 Application and Use

In the rapidly evolving digital landscape, government agencies and enterprises face increasing demands for efficiency, security, and innovation. As AI and machine learning become integral to operations, organizations must modernize their IT infrastructure to remain competitive and effective. The expansion of AI-driven initiatives requires a robust, flexible, and scalable foundation that supports advanced computing workloads while maintaining cost-effectiveness and operational efficiency. Opportunities to enhance IT operations through automation, predictive analytics, and modular data center solutions provide a pathway to achieving these goals, ensuring that agencies can securely and efficiently manage their technology resources.

Challenges: Implementing AI in government operations is not without its challenges. One of the primary challenges is the integration of AI at every scale with existing legacy systems that can be complex and costly. Additionally, there is a need for a skilled workforce that is proficient in AI technologies, which requires substantial investment in training and development. Data privacy and security are also major concerns, as AI systems often require access to large amounts of sensitive data.

Opportunities: To capitalize on these opportunities, agencies can implement IT automation strategies that streamline routine tasks, enabling remote management of critical infrastructure. Additionally, upgrading data centers to accommodate GPU-based servers will optimize AI-driven workloads, while advanced cooling and power management systems will enhance energy efficiency. Investing in modular data centers will further support evolving computational needs, allowing agencies to scale AI capabilities without excessive infrastructure costs. AI-driven predictive maintenance models can help reduce downtime and extend the lifespan of critical hardware, ensuring uninterrupted service delivery. Moreover, adopting on-premises AI clusters will provide sovereign control over sensitive data, reinforcing cybersecurity and compliance. Finally, fostering an AI-ready workforce through dedicated AI resources, training, and sandbox environments will empower government agencies to drive innovation and fully leverage AI's transformative potential.

Recommended actions include:

- Implementing AI operations to automate routine IT tasks and enable remote management of IT infrastructure, reducing the need for on-site personnel and ensuring efficient maintenance.
- Upgrading data centers to support GPU-based servers and implementing machine learning and automated control systems to optimize power and cooling needs, reducing energy consumption, and enhancing efficiency.
- Investing in modular data centers to meet the evolving needs of government agencies, ensuring support for advanced AI workloads without significant infrastructure investments.
- Utilizing AI-driven predictive maintenance models to reduce downtime, lower maintenance costs, extend equipment lifespan, and increase productivity by forecasting potential failures before they occur.
- Cutting the total cost of ownership of AI clusters by leveraging on-premises solutions and providing sovereign control over data and compute resources within the physical security domain.
- Developing agency-wide AI resources, including access to AI tools, training programs, and a sandbox environment for developing and testing AI applications, to foster an AI-ready workforce and drive innovation.
- Leveraging the tens of thousands of experts employed by the U.S. national laboratories, across physics, chemistry, biology, materials science, and computer science, including nearly one-third of all Ph.D.-level scientists and engineers in the country. This unparalleled talent pool is vital for AI advancements in science and engineering.

2.5.2 Cybersecurity

The cyberthreat landscape facing the United States is dominated by increasingly sophisticated, organized, industrialized malicious nation states and cybercriminal actors. These actors are using AI as a powerful tool to create more effective attacks, such as more realistic, plausible spear phishing and social engineering attacks. The threat spans the entire life cycle of hardware, firmware, software and data: supply chain, production, and operation. Mitigating this requires a multi-faceted defense-in-depth approach that also spans the entire life cycle to deliver AI systems that are secure by design, while delivering great performance and are easy to use.

Challenges: The possibility of a malicious actor having access to a cryptographically relevant quantum computer is a serious threat to the United States. Such a computer renders the current deployed secure protocols, firmware, and software verification mechanisms to be ineffective. Quantum-secure cryptographic algorithms exist, and, in many cases, it will be possible to upgrade software and firmware stacks in the field. Unfortunately, silicon parts cannot be changed after manufacturing, so it is urgent to transition hardware roots of trust and keys fused in silicon to quantum-secure algorithms. Failure to do so will mean that attackers

with access to a quantum computer will be able to sign malicious firmware that will be treated as legitimate by the system under attack, giving the threat actor full control over that system.

Opportunities: AI technologies can transform cybersecurity vulnerability management. The scarcity of trained, experienced cybersecurity professionals is a well-known challenge. While there has been a push towards automation for years, many tasks have remained manual. AI may assist by automating tasks, such as false positive filtering, that today generally rely on a human. AI may enable a level of simulating attacks, control effectiveness, and response capabilities far beyond what can be presently achieved. One especially compelling aspect of this potential for AI is to analyze complex attack chain scenarios that are currently intractable.

Recommended actions include:

- **Investing in protection against tampering of AI models and data**, such as trojan backdoors inserted by malicious actors in hardware, firmware, and software. This is particularly challenging for the open-source community where over-worked software maintainers are hard-pressed to provide adequate reviews of new code submitted to projects. AI-driven inspection together with cryptographic attestation and tracking of lineage can help mitigate this problem. Investment is required to develop these capabilities.
- **Investing in services for easy measurement and verification of operating platforms.** Although the basic concepts and hardware support for secure measurement and attestation have been in place for well over a decade (e.g., Trusted Platform Modules), the software and management services to enable their widespread use are missing.
- **Promoting industry-standard Trusted Execution Environments (TEE)** for processors, GPUs, and other devices. TEEs have given rise to an emerging paradigm called Confidential Computing in which data is protected by encryption at rest, in transit, and in memory. Data is accessible only to verified TEEs; it is isolated by encryption from all other entities. This offers greatly enhanced protection of data from malicious insiders and from malicious actors who have stolen credentials from legitimate operators.

2.5.3 Data Privacy and Security

Robust data and model provenance and governance enable AI pipelines and collaboration.

Challenges: AI systems have attack surfaces which are different from non-AI systems. In the supply chain, this includes model tampering, poisoning, and data tampering. In operation, the difficulty in bounding the behavior of AI systems makes it challenging to secure applications and introduces new attacks such as prompt injection, jail breaking, perturbation, model reprogramming, hallucinations, and model stealing. For safe operation, further investment is required to protect against these new attacks.

Opportunities: AI systems have a significant role to play in securing the operation of other AI systems and a robust data and model governance framework is required that prioritizes data and model quality, security, privacy, and accessibility.

Recommended actions include:

- **Establishing guidelines for clear ownership and licensing, rigorous quality assurance,** and strong security measures. Tracking data and model lineage, and ensuring user-friendly access maximizes the data value and explainability of derived AI products.
- **Investing in a “knowledge nurturing platform” to accelerate AI in scientific discovery,** innovation, and enhance public trust. It should include a collaborative platform, a data historian, a data vault, easy and fluid data services, incentive models for both contributors and curators, and embargoed access policy enforcement tools.
- **Establishing robust data and metadata governance practices** that are key for the success of AI-based pipelines for science, given the tight dependence of AI models on data.

2.5.4 National Security and Defense

Technology is at a tipping point in the age of AI. AI promises advancements at all levels of computing and has the potential to revolutionize science and engineering. By integrating AI with modeling, simulations, big data, and high-performance data analytics, the Department of Defense (DoD) and the DOE’s National Nuclear Security Administration (NNSA) can dramatically increase the value of advanced computing for the national security community and the nuclear security enterprise. As AI transforms the national security landscape, high-performance computing will become an even more essential tool for the defense of our nation. The rapid evolution of AI demands immediate and strategic action to maintain technological superiority and battlefield dominance.

Challenges: The integration of AI into national security presents several challenges, such as in *Infrastructure and Investment* where building state-of-the-art AI systems requires substantial computational resources, massive datasets, and highly specialized expertise, necessitating significant government investment. Regarding *Security Risks*, the DoD must ensure classified national security data, AI models, and AI artifacts remain secure, responsibly managed, and safely implemented to prevent adversarial exploitation. On *Counter-Adversarial AI*, the development of AI-driven countermeasures is essential to predict, prevent, and mitigate threats to critical infrastructure, energy security, nuclear nonproliferation, biological, chemical safety, and cybersecurity. Related to *Workforce Readiness*, scientists and engineers must be equipped with AI expertise as AI increasingly becomes indispensable for defense technology.

Opportunities: To maintain technological superiority, the U.S. Government can leverage its existing strengths in computing infrastructure, ecosystems, partnerships, and expertise in the several ways. *Leveraging the DOE National Labs’* world-class supercomputing ecosystems will allow for managing AI data at every scale and facilitating the development and deployment of classified and unclassified AI platforms, tools, and applications. With *Expanding High-Performance Computing Capabilities*, the DoD’s High-Performance Computing Modernization Program (HPCMP) already houses some of the world’s most advanced supercomputing resources. Additional investments can enhance the ability to support AI-driven defense innovations. The Fiscal Year 2025 National Defense Authorization Act (NDAA) calls for the expansion of high-performance computing infrastructure to support next-generation AI

capabilities. By *Developing AI Systems and Solutions for Defense*, AI can improve predictive capabilities, expand design possibilities, and reduce costs in weapons system development, simulations, threat assessments, and other areas important to national defense.

Recommended actions include:

- **Expanding High-Performance Computing Investments for AI:** Prioritize funding for programs like the HPCMP to ensure DoD scientists and engineers have access to the most advanced AI computing infrastructure for the science and technology, acquisition engineering, and testing and evaluation communities.
- **Developing Secure AI Models for National Security:** Leverage classified and secure unclassified data to develop AI models tailored for sensitive defense applications, ensuring secure and responsible deployment.
- **Enhancing Public-Private Collaboration:** Strengthen partnerships between the DoD, DOE, industry, and academia to accelerate AI advancements and integrate the latest innovations into defense applications.
- **Fortifying AI Trust and Security:** Utilize U.S. Government expertise in evaluating dual-use technologies to assess vulnerabilities, conduct red-teaming, and implement rigorous threat testing of up to frontier scale AI models.
- **Accelerating AI Workforce Development:** Ensure that military and defense personnel, as well as DoD scientists and engineers, are trained to effectively develop and deploy AI technologies in support of national security missions.

2.5.5 Research and Development

The United States Government's federal scientific and national security missions demand world-leading AI capabilities. AI has already demonstrated its potential in scientific breakthroughs across diverse fields, including quantum chemistry, cancer cell behavior, molecular dynamics, and bioenergy. To build on these advancements, a transformational effort is required to establish a nation-scale AI ecosystem. This effort must be deliberate, coordinated, and supported by substantial, sustained government funding and support.

Challenges: Building an advanced AI research and development capability requires significant investments in infrastructure, data management, and responsible AI practices. This includes the need for dedicated compute platforms to support AI model development, testing, and deployment. Also, it requires ensuring AI-ready scientific data is available at scale while maintaining security and accessibility and addressing security vulnerabilities and dual-use risks in AI models to ensure responsible and safe deployment. Finally, enhancing collaboration across federal agencies, academia, and industry while maintaining national interests and data security will be essential.

Opportunities: To achieve AI leadership, the U.S. Government can leverage its existing strengths in computing infrastructure, data generation, and workforce expertise. *Dedicated Compute Platforms* can be acquired and deployed, including scaled testbeds to co-design next-generation AI hardware, modular systems for training smaller models, and full-scale

infrastructure for large foundation model training and post-training. *AI-Ready Scientific Data* can transform the U.S. Government's vast scientific datasets into structured "data lakes" to support multi-domain foundation models and advance multiple scientific fields. *Responsible AI Research* should focus on establishing foundational research initiatives to assess AI model security, conduct red-teaming, and develop trustworthy AI algorithms. *Public-Private Partnerships* can be formed that strengthen collaborations between government agencies, academia, and industry to accelerate and make use of AI advancements.

Recommended actions include:

- **Establishing a Research and Development Program Focused on Four Pillars:**
 - **Data:** Developing tools and protocols for efficient AI dataset aggregation, curation, and secure distribution.
 - **Computing Infrastructure:** Investing in next-generation AI testbeds, energy-efficient AI hardware, and high-performance computing systems.
 - **AI Models and Systems:** Building and validating AI foundation models for scientific and national security applications.
 - **Application:** Developing tailored AI solutions to address pressing scientific, energy, and security challenges.
- **Developing a Federal AI Initiative that includes:**
 - Establishing a multi-year public-private partnership program to drive post-exascale AI computing technology development.
 - Aligning AI investments across federal agencies to ensure strategic coordination and resource efficiency. Federal AI research hubs could provide incentive for commercialization of innovative R&D projects across industry and academia.
 - Ensuring classified AI models remain secure, responsibly managed, and aligned with national security interests.
- **Enhancing Federal Coordination and Investment by:**
 - Strengthening inter-agency AI R&D collaboration while maintaining safeguards against external influence.
 - Providing multi-year funding mechanisms to ensure sustained AI advancements.
 - Establishing secure facilities for AI research to develop models trained on unique federal datasets unavailable to the private sector.

2.5.6 Innovation and Competition

AI is a major driver of innovation across various industries, including healthcare, finance, and transportation. The competitive landscape is rapidly evolving, with new startups and established tech giants driving advancements. It is important to maintain a competitive edge in AI by leveraging open standards and hybrid computing solutions. That includes fostering

development of AI models that are not solely dependent on proprietary cloud APIs, which can cause vendor lock-in and hinder innovation.

Challenges: Ensuring a level playing field and preventing unfair competitive practices and balancing innovation with ethical considerations and regulatory compliance.

Opportunities: Encouraging competition through supportive policies and investments in AI research and development. The use of standardized, hardware-agnostic programming models can enable scalability and flexibility across different computing environments. This approach ensures that developers can move their AI workloads seamlessly between on-premises HPC systems and cloud resources, thereby avoiding the pitfalls of being tied to a single provider's ecosystem. The role of leadership-class HPC supercomputing centers is key in supporting U.S. AI development in domains like science, security, and engineering. These centers provide the necessary infrastructure for training large-scale AI models, which often require significant computational power that goes beyond the capabilities of typical cloud environments. By investing in HPC systems optimized for a combination of traditional HPC and AI workloads, the U.S. Government can support the unique computational and data requirements of scientific and engineering applications. Growing the investment in leadership systems can integrate AI-enhanced scientific workloads, using energy-efficient processors and advanced cooling systems. This not only supports cutting-edge research but also ensures that the U.S. remains at the forefront of AI innovation.

Recommended actions include:

- Growing the investments in leadership class supercomputers.
- Promoting open standards, including establishing a clear intellectual property and copyright framework, to reduce proprietary lock-in and AI monopolization.
- Investing in public-private partnerships to drive innovation.

2.5.7 Procurement

Supercomputers are essential for training large-scale AI models. The procurement process involves balancing performance, speed of deployment, cost, and energy efficiency. To address the rapid adoption of new technologies, the procurement process for new government-funded AI supercomputers should be significantly restructured to accommodate the fast-paced advancements in AI and related hardware. The implementation of a structured framework contract could support incremental deliveries and facilitate quick early deployment of technologies and provide faster access to new advancements without the overhead of individual procurements. By allowing for incremental upgrade points within the contract period, agencies like the DOE can improve upon the traditional five-year cycle of deploying a single leadership-class machine, thus better matching the current pace of technological change.

Challenges include high costs of AI systems, the rapid obsolescence of hardware, and ensuring that procurement processes support cutting-edge research without stifling innovation.

Opportunities include leveraging public-private partnerships to share resources and costs and investing in energy-efficient supercomputing technologies. Also, a flexible procurement

strategy can accommodate continuous integration, deployment, and iterative prototyping and development of new feature sets. This involves collaborative efforts with industry partners to ensure that the procurement process is not only streamlined but also responsive to the evolving requirements of AI workloads. By incorporating these recommendations, the U.S. Government can enhance its ability to quickly deploy cutting-edge technologies, thereby maintaining its leadership in AI and high-performance computing.

Recommended actions include:

- Simplifying procurement processes to allow for rapid deployment of new technologies.
- Supporting incremental upgrades to keep pace with technological advancements.
- Fostering partnerships to share supercomputing resources and expertise.

2.6 Global Leadership

Continued U.S. leadership in AI requires a coordinated approach to ensuring that U.S. technology and standards remain the global benchmark. Global market access and growth are central to U.S. companies' ongoing ability to invest in R&D to maintain technological leadership and continue driving U.S.-led innovation. Further, international collaboration is crucial for advancing AI research and addressing global challenges. Initiatives such as the Global Partnership on AI (GPAI) and Trillion Parameter Consortium (TPC) promote cross-border cooperation.

Challenges include navigating geopolitical tensions and ensuring equitable participation in the advancement of AI and addressing data sovereignty and privacy concerns.

Opportunities include strengthening international partnerships to share knowledge, resources, and best practices. Also, international collaboration on AI, particularly through initiatives like the TPC, provides a collaborative platform for research teams worldwide to solve significant AI research challenges, particularly those relevant to science. The aim is to create a focal point for competing with commercial AI initiatives that are driven by substantial business interests. By making models openly available, the TPC encourages scientists to build upon each other's work, leading to rapid advancements and breakthroughs. This collaborative environment fosters interdisciplinary research where insights from different fields converge to address complex scientific problems. Finally, open sourcing promotes transparency, which is crucial for scientific integrity and reproducibility. Researchers can inspect models, understand their workings, and verify results, contributing to more responsible and trustworthy AI systems.

Recommended actions include:

- Ensuring foreign markets are open to U.S. AI technology.
- Promoting international AI research collaborations.
- Establishing frameworks for data sharing that respect privacy and sovereignty.
- Encouraging global adoption of transparent and ethical AI standards.



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